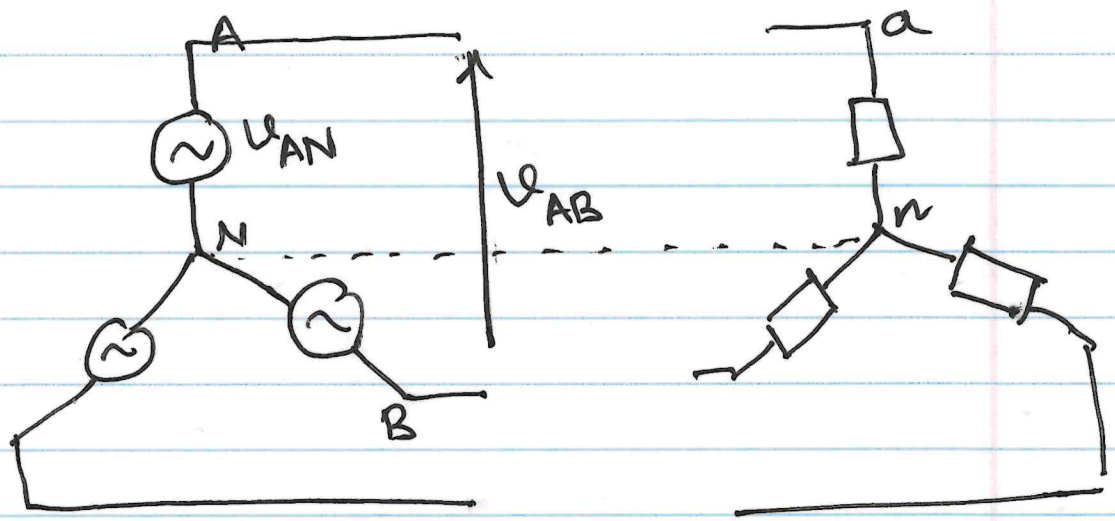
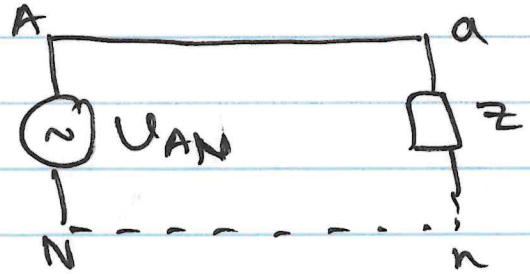
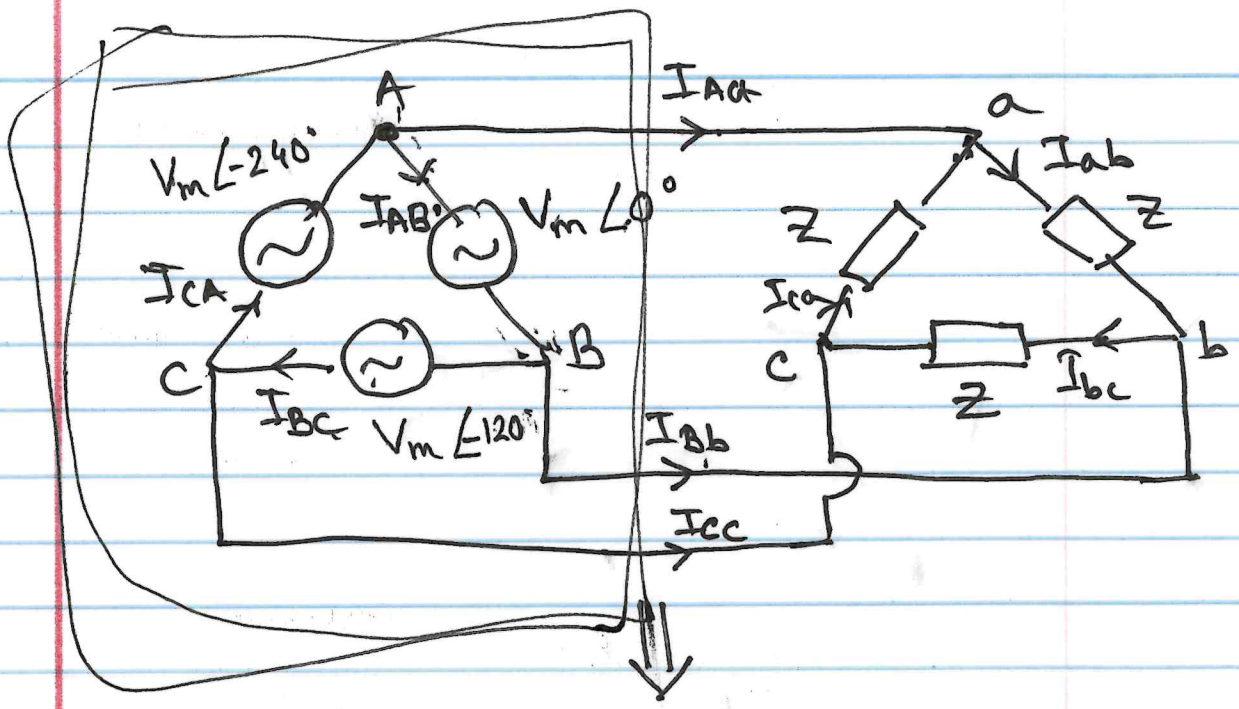
Lecture 6 ECE-330.

Δ/Y Δ/Y







Line Qty : Source

V_{AB}, V_{BC}, V_{CA}

I_{Aa}, I_{Bb}, I_{cc}

Load

V_{ab}, V_{bc}, V_{ca}

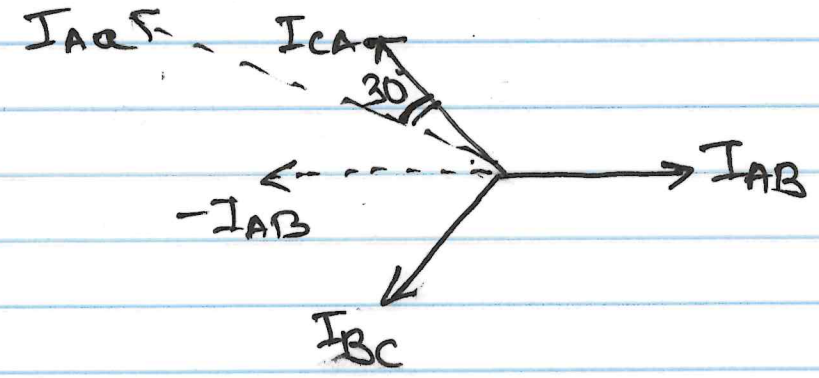
I_{Aa}, I_{Bb}, I_{cc}

Phase Qty : V_{AB}, V_{BC}, V_{CA}

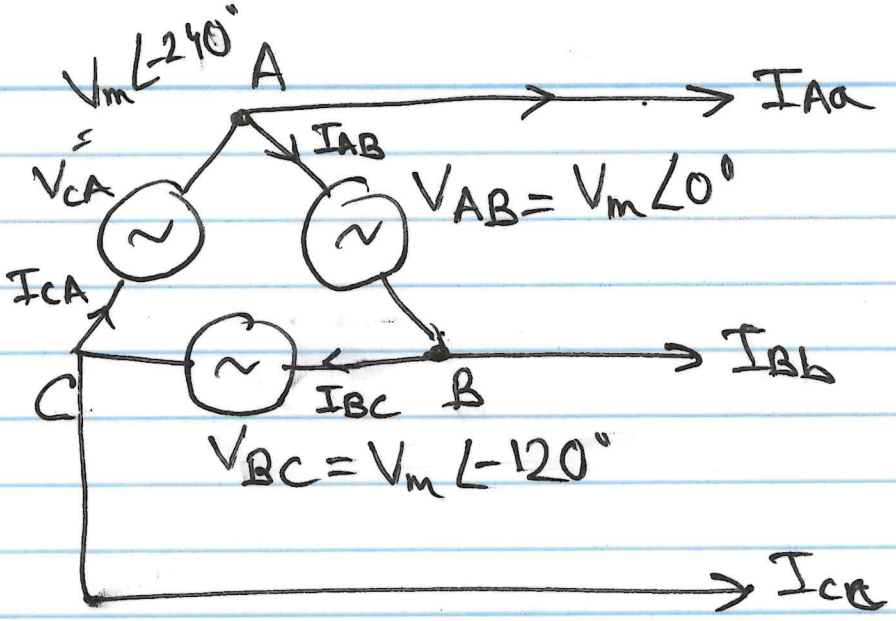
I_{AB}, I_{BC}, I_{CA}

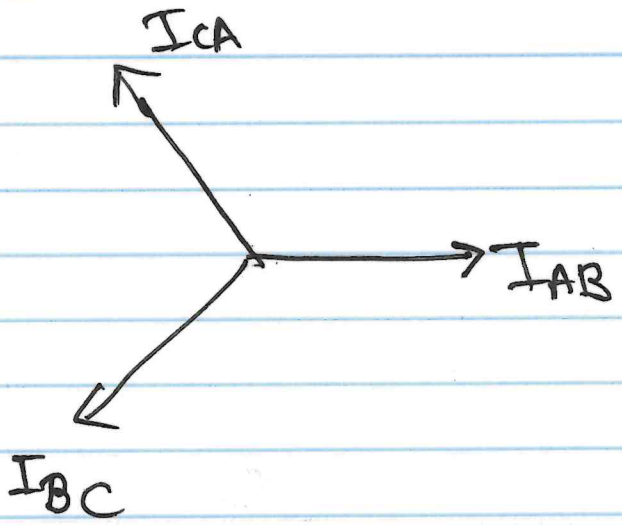
V_{ab}, V_{bc}, V_{ca}

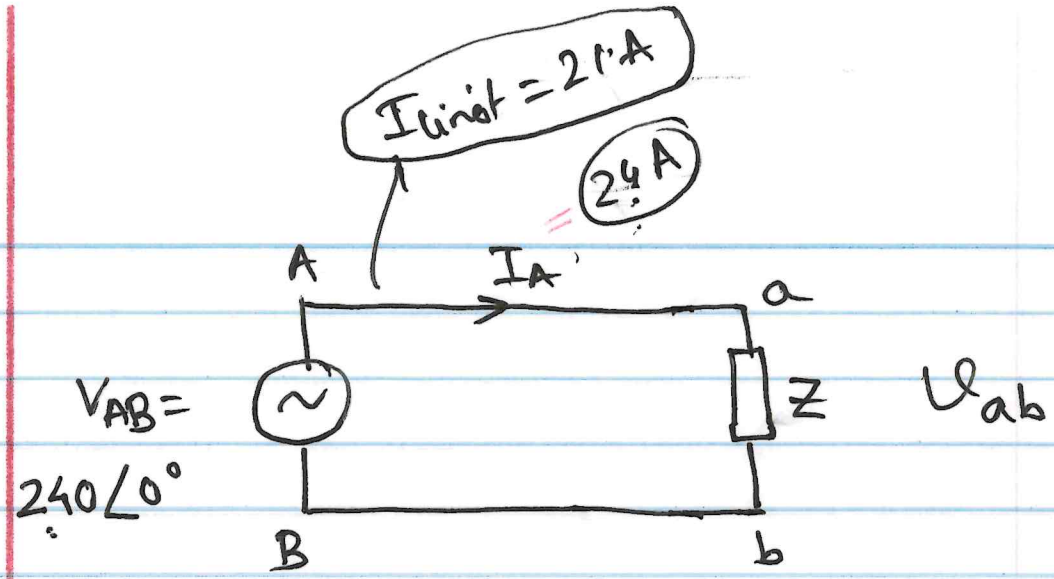
I_{ab}, I_{bc}, I_{ca}



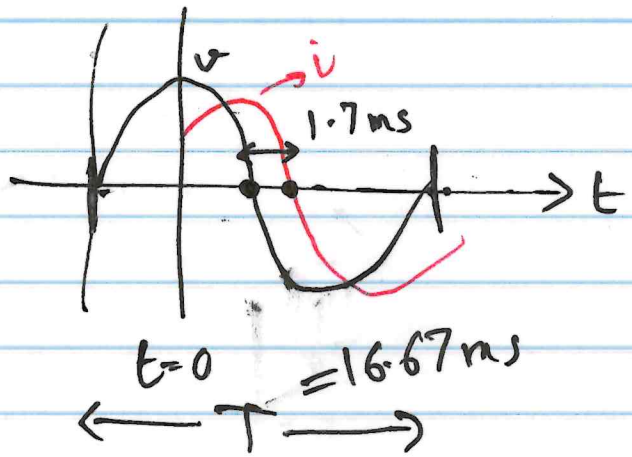
$I_{AA'} = I_{CA} - I_{AB}$


$V_m \angle -240°$
A
I_{AA}
I_{AB}
V_{CA}
$V_{AB} = V_m \angle 0°$
I_{CA}
I_{BB}
C
I_{BC}
B
$V_{BC} = V_m \angle -120°$
I_{CC}


I_{CA}
I_{AB}
I_{BC}



$|I_A| = 24$ A rms.



ϕ = angle between V and I

16.67 ms $\equiv 360°$

1.7 ms $\quad 36.72°$

$\bar{V}_{AB} = 240\angle 0°$, $\bar{I}_A = 24\angle -36.72°$

$Z = \dfrac{\bar{V}_{AB}}{\bar{I}_A}$ $\quad$ $\bar{V}_{AB}$, $-36.72°$, $\bar{I}_A$

R-L load

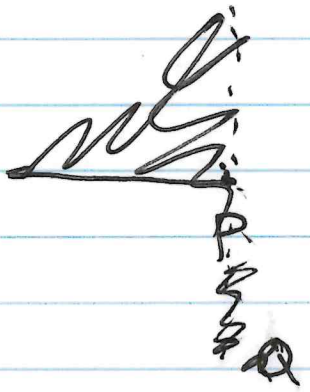
$= 10\angle 36.72\ \Omega = 10\cos 36.72 + j\,10\sin 36.72$

$(8 + j6)\ \Omega$

S, P, Q

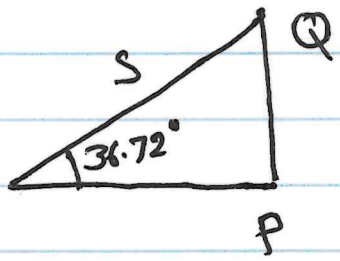
$P = V_{rms} I_{rms} \cos\phi$

$Q = V_{rms} I_{rms} \sin\phi$



Q is positive → RL load

Q is negative → RC load.

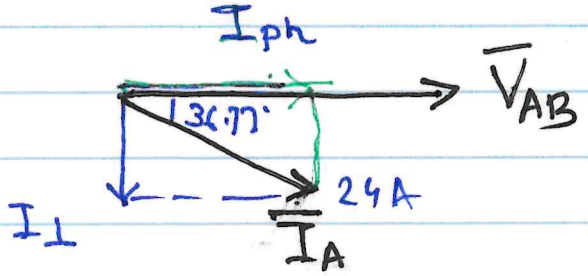


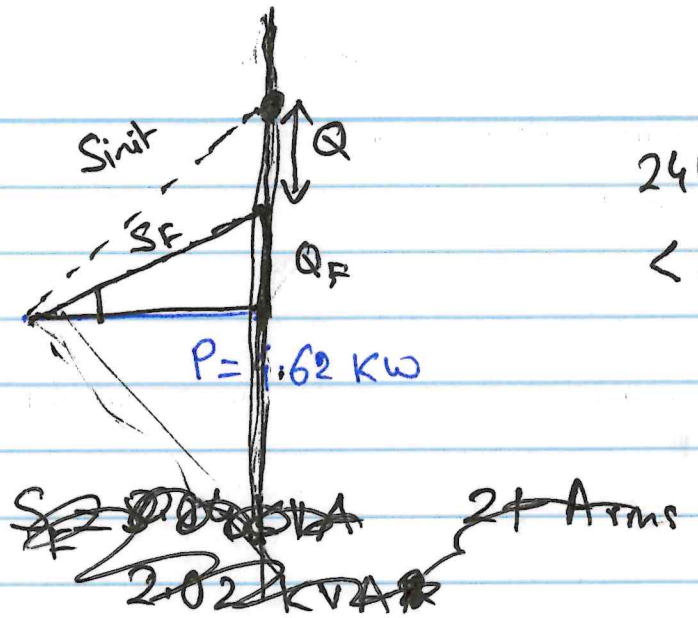
$P^2 + Q^2 = S^2$

$S = V_{rms} I_{rms} = 240 \times 24$ VA

$P = V_{rms} I_{rms} \cos\phi = 4.62$ KW

$Q = V_{rms} I_{rms} \sin\phi = 3.44$ KVAR





240 x 21

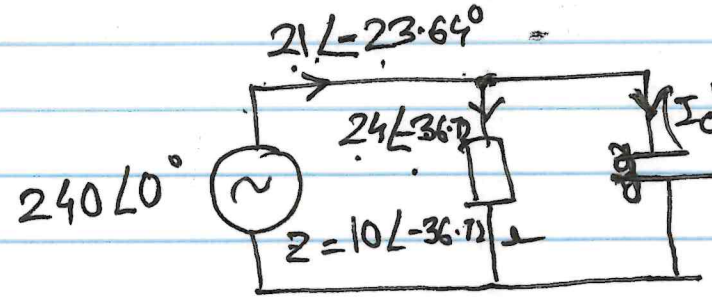
< 240 x 24

$S_{init} = 5.76$ KVA | $Q_{init} = 3.44$ KVAR

$S_F = 5.04$ KVA | $Q_F = 2.01$ KVAR

$$\sqrt{5.04^2 - 4.62^2}$$

Q supplied locally = $(3.44 - 2.01)$ KVAR



$$|I_C| = \left|\frac{240}{X_C}\right|$$

$$X_C = \left(\frac{240}{|I_C|}\right)$$

$C = 65.5\ \mu F$

$\frac{1}{\omega C} =$